

VALUE ADDITION COURSE

Title of Certification Course: Generative AI

Purpose of Course: To empower individuals with practical skills in generative AI, enabling them to create text, images, audio, video, and animations using free and accessible tools. Participants will learn how to leverage AI for creativity, productivity, and innovation.

Course Objective: To provide foundational knowledge of generative AI concepts. To enable participants to use AI tools effectively for various creative tasks. To explore ethical considerations and responsible AI usage

Course Content

Unit 1: Introduction to Generative AI

- 1.1 Understanding Generative AI and its applications.
- 1.2 Overview of free tools for text, image, audio, video, and animation generation.
- 1.3 Ethical considerations and responsible usage of AI tools.

Unit 2: AI for Text Generation

- 2.1 Generating content for blogs, emails, and storytelling using AI.
- 2.2 Exploring chatbot creation basics.
- 2.3 Using generative AI for brainstorming and ideation.

Unit 3: AI for Image Generation

- 3.1 Basics of text-to-image AI models.
- 3.2 Creating art, illustrations, and visuals using prompts.
- 3.3 Enhancing existing images with AI tools.

Unit 4: AI for Audio Generation

- 4.1 Basics of audio generation and voice modulation.
- 4.2 Creating podcasts, voiceovers, and background music.
- 4.3 Enhancing audio quality using AI.

Unit 5: AI for Video & Animation Generation

- 5.1 Generating short videos with AI.
- 5.2 Creating animations and visual storytelling.
- 5.3 Using AI for enhancing video production workflows.

Unit 6: Capstone Project and Showcase

- 6.1 Participants create a project integrating text, image, audio, video, or animation generation.
- 6.2 Showcase and peer feedback on projects.
- 6.3 Group discussion on future trends and individual applications of AI tools

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MCQ BANK – Generative AI

Unit 1: Introduction to Generative AI

1. Generative AI differs from traditional AI mainly because it:
 - a) Classifies data only
 - b) Generates new data resembling training data
 - c) Only predicts future outcomes
 - d) Works without any dataset

Answer: b

2. Which of the following is an application of Generative AI?
 - a) Predicting stock prices
 - b) Detecting spam emails
 - c) Creating synthetic images
 - d) Sorting emails into folders

Answer: c

3. Tools like ChatGPT and Jasper primarily belong to which category?
 - a) Image generation
 - b) Text generation
 - c) Audio generation
 - d) Animation generation

Answer: b

4. DALL·E, Stable Diffusion, and MidJourney are primarily used for:
 - a) Speech-to-text
 - b) Text-to-image generation
 - c) Video summarization
 - d) Chatbot creation

Answer: b

5. Which is a key ethical concern in generative AI?
 - a) Data labeling
 - b) Overfitting
 - c) Deepfakes and misinformation
 - d) Dataset normalization

Answer: c

6. Free tool “Runway ML” can generate which of the following?
 - a) Blog posts
 - b) Images and videos
 - c) Email templates
 - d) Chatbot workflows

Answer: b

7. Which of these is NOT a generative AI domain?
 - a) Image creation
 - b) Text summarization
 - c) Weather prediction
 - d) Audio synthesis

Answer: c

8. The term "responsible AI usage" primarily emphasizes:
 - a) Reducing GPU cost

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- b) Following ethical guidelines
- c) Optimizing datasets
- d) Speeding up training

Answer: b

9. A potential misuse of generative AI is:

- a) Automatic note-taking
- b) Producing realistic fake news
- c) Brainstorming new ideas
- d) Creating animations

Answer: b

10. Which of the following best describes "hallucination" in AI tools like ChatGPT?

- a) Tool stops responding
- b) Tool generates incorrect but confident outputs
- c) Tool repeats answers
- d) Tool fails to generate outputs

Answer: b

Unit 2: AI for Text Generation

11. AI-based blog generation often uses which model type?

- a) Convolutional Neural Networks
- b) Transformer-based models
- c) Decision Trees
- d) K-means clustering

Answer: b

12. Which platform allows creating chatbots without coding using generative AI?

- a) Canva
- b) ManyChat
- c) Blender
- d) Photoshop

Answer: b

13. Storytelling using AI can be enhanced by:

- a) Restricting model creativity
- b) Using structured prompts
- c) Providing no input
- d) Avoiding feedback loops

Answer: b

14. Which of these is NOT an application of AI in text generation?

- a) Drafting emails
- b) Writing novels
- c) Image synthesis
- d) Resume writing

Answer: c

15. AI brainstorming tools are useful for:

- a) Generating raw ideas
- b) Editing videos
- c) Correcting grammar only

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d) Designing logos

Answer: a

16. Chatbot creation basics usually require:

- a) Domain-specific knowledge base
- b) 3D rendering engines
- c) Audio synthesizers
- d) Blockchain network

Answer: a

17. A common limitation of AI-generated content is:

- a) High storage requirement
- b) Lack of factual accuracy
- c) Slow processing speed
- d) Requirement of big data centers

Answer: b

18. Which AI tool is known for email content generation?

- a) Jasper AI
- b) DALL·E
- c) Runway ML
- d) Stable Diffusion

Answer: a

19. The process of fine-tuning a chatbot refers to:

- a) Hardware optimization
- b) Training model on domain-specific data
- c) Reducing input length
- d) Adding more parameters randomly

Answer: b

20. Generative AI used in ideation can:

- a) Automate debugging
- b) Suggest creative campaign strategies
- c) Forecast financial growth
- d) Compress large datasets

Answer: b

Unit 3: AI for Image Generation

21. The primary input to text-to-image AI models is:

- a) Sample image
- b) Text prompt
- c) Audio signal
- d) Video clip

Answer: b

22. Which algorithm family powers most text-to-image tools?

- a) Reinforcement learning
- b) Diffusion models
- c) Decision trees
- d) Clustering algorithms

Answer: b

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23. Stable Diffusion is mainly known for:

- a) Email generation
- b) High-resolution image creation
- c) Chatbot scripting
- d) Voice cloning

Answer: b

24. Which is a typical use of AI-generated visuals?

- a) Content illustration
- b) Statistical computation
- c) Hardware testing
- d) Cloud storage

Answer: a

25. Which factor most affects image output quality in generative AI?

- a) Prompt engineering
- b) Audio filters
- c) Dataset compression
- d) Email template design

Answer: a

26. Enhancing existing images using AI is called:

- a) Image inpainting or upscaling
- b) Prompt filtering
- c) Batch normalization
- d) Tokenization

Answer: a

27. Which tool allows AI-powered art generation?

- a) Canva Magic Media
- b) Gmail
- c) Zoom
- d) Google Sheets

Answer: a

28. A challenge in AI-generated images is:

- a) Incorrect object proportions
- b) Faster rendering times
- c) Overfitting chatbot data
- d) Reduced text summarization

Answer: a

29. MidJourney primarily works through:

- a) Standalone app
- b) Discord server interface
- c) Browser extensions only
- d) Spreadsheet plugins

Answer: b

30. Which concept helps improve visual storytelling with AI images?

- a) Style consistency
- b) Random prompt variation
- c) Noise injection only

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d) Reducing text length

Answer: a

Unit 4: AI for Audio Generation

31. Voice modulation in AI audio is achieved through:

- a) Pitch and tone alteration
- b) Dataset compression
- c) Image filters
- d) Prompt length

Answer: a

32. Which is an example of AI-based voiceover tool?

- a) Synthesia
- b) MidJourney
- c) DALL·E
- d) Canva Docs

Answer: a

33. Podcasts created using AI tools require:

- a) Speech synthesis + editing tools
- b) Image diffusion models
- c) Database queries
- d) Blockchain contracts

Answer: a

34. AI audio can generate:

- a) Voiceovers for e-learning
- b) Code compilers
- c) Sorting algorithms
- d) Document versioning

Answer: a

35. Enhancing audio quality often involves:

- a) Noise reduction and equalization
- b) Prompt tokenization
- c) Storyline structuring
- d) Image masking

Answer: a

36. Which tool is widely used for AI-generated background music?

- a) Amper Music
- b) Photoshop
- c) Blender
- d) Excel

Answer: a

37. A challenge in AI-generated audio is:

- a) Robotic or unnatural voice output
- b) Lack of GPU for training
- c) Excessive storytelling
- d) Limited chatbot context

Answer: a

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38. AI-driven transcription tools like Otter.ai belong to:

- a) Audio-to-text category
- b) Image-to-text category
- c) Video-to-animation category
- d) Text-to-video category

Answer: a

39. Which neural network type is often used in audio synthesis?

- a) Recurrent Neural Networks
- b) Decision Trees
- c) Random Forests
- d) Logistic Regression

Answer: a

40. Using AI for multilingual voice generation helps in:

- a) Global content accessibility
- b) Faster rendering only
- c) Preventing hallucinations
- d) Reducing email spam

Answer: a

Unit 5: AI for Video & Animation Generation

41. Which tool allows AI-based avatar video creation?

- a) Synthesia
- b) Gmail
- c) Canva Docs
- d) Excel

Answer: a

42. Text-to-video AI models generate:

- a) Animations or video clips from textual descriptions
- b) Chatbot responses only
- c) Podcasts
- d) Emails

Answer: a

43. Which is an advantage of AI in video production workflows?

- a) Automated editing and subtitles
- b) GPU heating issues
- c) Data labeling requirement
- d) Fewer prompts needed

Answer: a

44. AI-enhanced animation tools can help in:

- a) Character lip-syncing
- b) Spreadsheet calculations
- c) Noise equalization
- d) Prompt tokenization

Answer: a

45. Runway ML is often used for:

- a) AI-based video editing and generation
- b) Writing resumes

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- c) Creating email templates
- d) Designing presentations only

Answer: a

46. Which of the following is a limitation of AI-generated video?

- a) Unrealistic human movements
- b) Faster editing speeds
- c) Subtitle automation
- d) Enhanced transitions

Answer: a

47. Visual storytelling using AI is enhanced by:

- a) Consistency across frames
- b) Random variation in style
- c) Eliminating all prompts
- d) Removing soundtracks

Answer: a

48. A practical use case of AI video generation is:

- a) Training simulations
- b) Sorting numbers
- c) Grammar checking
- d) Image diffusion

Answer: a

49. Which factor influences quality of AI video outputs most?

- a) Input prompts + dataset diversity
- b) Noise filters only
- c) Faster compilation
- d) Cloud storage type

Answer: a

50. Future trends in AI video generation are likely to include:

- a) Real-time personalized content creation
- b) Elimination of visual data
- c) Replacing text generation
- d) Limiting video creativity

Answer: a

Unit 6: Capstone Project and Showcase:

51. The main objective of a Capstone Project in Generative AI is to:

- a) Learn basic programming syntax
- b) Integrate multiple AI modalities into a practical application
- c) Perform only theoretical research
- d) Focus solely on ethical guidelines

Answer: b

52. Peer feedback during a project showcase is primarily useful for:

- a) Reducing hardware costs
- b) Improving project quality through collaborative insights
- c) Avoiding the use of datasets
- d) Finalizing the future trends of AI

Answer: b

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53. Which of the following is an example of a multimodal AI capstone project?
- a) Chatbot generating FAQs
 - b) Blog summarization tool
 - c) Interactive storytelling with AI-generated text, visuals, and narration
 - d) Spreadsheet formula generator

Answer: c

54. Group discussions on future trends in AI tools help participants to:
- a) Replicate existing solutions without innovation
 - b) Identify emerging applications and adapt their skills
 - c) Avoid interdisciplinary approaches
 - d) Focus only on text-based outputs

Answer: b

55. An individual application of AI tools in professional settings could include:
- a) Creating deepfake misinformation
 - b) Automating repetitive content creation tasks
 - c) Replacing all human decision-making processes
 - d) Blocking collaboration opportunities

Answer: b

Short Questions — Generative AI (Units 1–3)

1. Define Generative AI and explain one way it differs from traditional discriminative AI.
2. Identify two practical applications of Generative AI in academic or professional contexts and explain briefly.
3. List any two free tools available for AI-based image or video generation and mention their primary functions.
4. Explain the role of ethical considerations when using AI for creating synthetic media.
5. What is meant by “responsible usage” of AI tools? Give one example in the context of text generation.
6. How can generative AI support the creation of blog posts and emails, and what is one risk associated with over-reliance on such tools?
7. State the basic principle of chatbot creation using generative AI.
8. Provide one example of how AI can assist in brainstorming or ideation for product design or research.
9. Compare the output of rule-based chatbots with that of AI-driven generative chatbots.
10. Explain the basic working idea of text-to-image models such as Stable Diffusion.
11. What is the significance of “prompt design” in generating high-quality images?
12. Give an example of how AI can enhance an existing image (e.g., inpainting, upscaling) and explain briefly.
13. Evaluate one potential misuse of AI image-generation tools in social media contexts.
14. Suggest one guideline that could help reduce bias or harmful content in AI-generated text or images.

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15. A researcher wants to generate illustrations for an educational e-book using free AI tools. Recommend one tool and justify your choice briefly.

Short Questions — Units 4, 5, and 6

Unit 4: AI for Audio Generation

16. Define voice modulation in the context of AI-driven audio generation and state one practical application.
17. Explain how neural networks can synthesize realistic human voices in audio generation.
18. Identify two challenges in creating natural-sounding AI-generated podcasts.
19. Compare AI-based voiceover generation with traditional studio recording in terms of efficiency and limitations.
20. List two AI techniques used to enhance audio quality, and briefly describe how one of them works.

Unit 5: AI for Video & Animation Generation

21. Explain the basic principle of text-to-video generation using AI.
22. Identify one AI tool for video creation and describe its key functionality.
23. How does AI facilitate the process of lip-syncing in animated characters?
24. Compare the use of AI in short video generation with AI in creating full-length animations.
25. What role does AI play in automating post-production video workflows? Provide one example.

Unit 6: Capstone Project and Showcase

26. Describe the purpose of a capstone project in the context of generative AI learning.
27. Suggest one way to integrate both audio and video generation into a single capstone project.
28. Why is peer feedback important during the showcase phase of an AI project?
29. Identify one potential ethical issue that could arise in a student's AI-generated multimedia project.
30. Propose one future trend in AI-based creative media that students might discuss during group reflections.